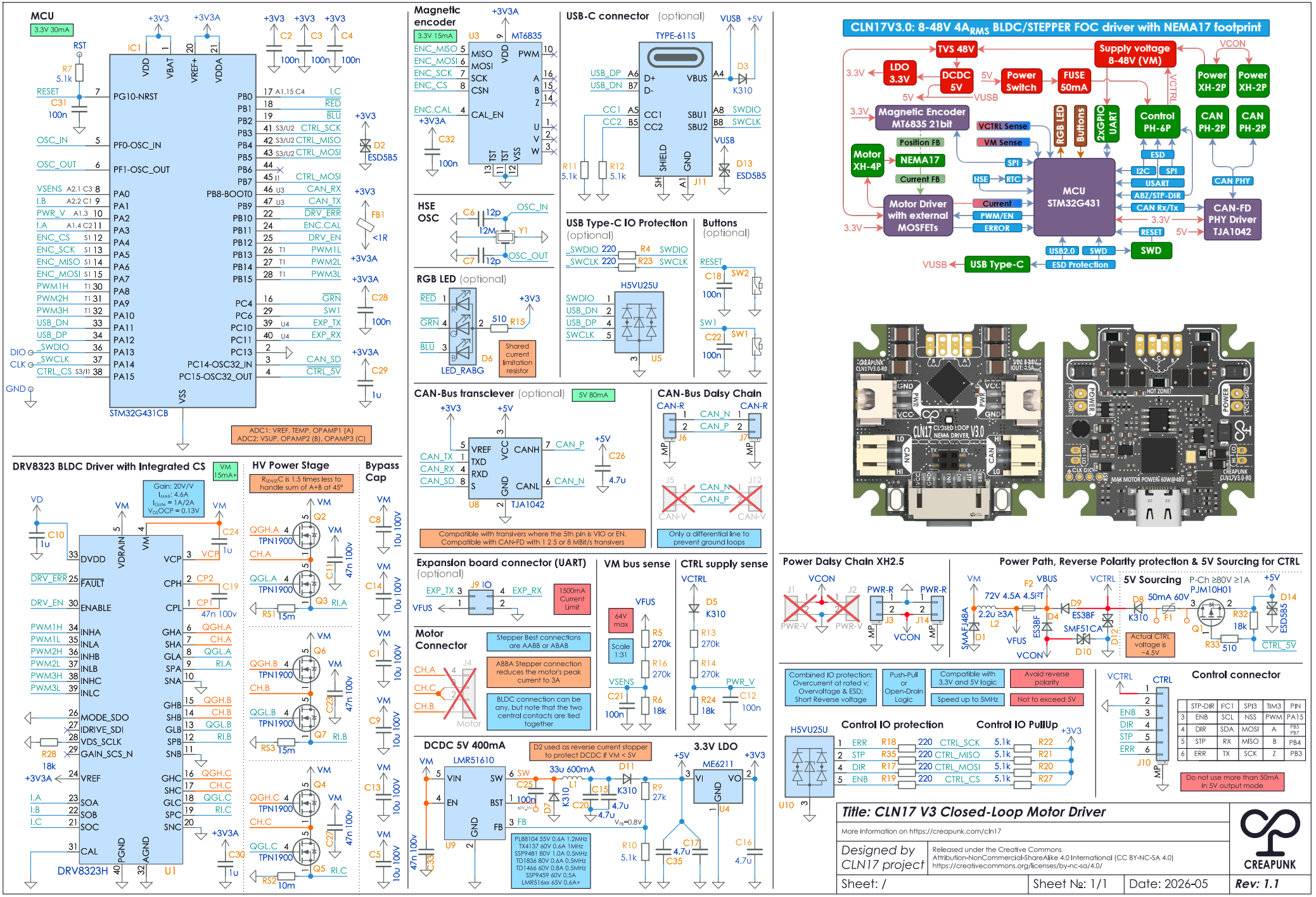

MCU
3.3V 30mA
STM32G431CB
ADC1: VREF, TEMP, OPAMP1 (A)
ADC2: VSUP, OPAMP2 (B), OPAMP3 (C)
Magnetic encoder
3.3V 15mA
MT6835
USB-C connector (optional)
TYPE-611S
HSE OSC
RGB LED (optional)
LED_RABG
Shared current limitation resistor
USB Type-C IO Protection (optional)
H5VU25U
Buttons (optional)
CAN-Bus transciever (optional)
5V 80mA
TJA1042
Compatible with transivers where the 5th pin is VIO or EN.
Compatible with CAN-FD with 1 2 5 or 8 MBit/s transivers
CAN-Bus Daisy Chain
Only a differential line to prevent ground loops
DRV8323 BLDC Driver with Integrated CS
VM 15mA+
Gain: 20V/V
I_MAX: 4.6A
I_SCMIN = 1A/2A
V_DSOCP = 0.13V
DRV8323H
HV Power Stage
R_SENSE C is 1.5 times less to handle sum of A+B at 45°
Bypass Cap
Expansion board connector (UART)
(optional)
1500mA Current Limit
VM bus sense
64V max
Scale 1:31
CTRL supply sense
Motor Connector
Stepper Best connections are AABB or ABAB
ABBA Stepper connection reduces the motor's peak current to 3A
BLDC connection can be any, but note that the two central contacts are tied together
DCDC 5V 400mA
LMR51610
D2 used as reverse current stopper to protect DCDC if VM < 5V
PL88104 55V 0.6A 1.2MHz
TX4137 60V 0.6A 1MHz
SSP9481 80V 1.0A 0.5MHz
TD1836 80V 0.6A 0.5MHz
TD1466 60V 0.8A 0.5MHz
SSP9459 60V 0.5A
LMR516xx 65V 0.6A+
3.3V LDO
ME6211
CLN17V3.0: 8-48V 4A_RMS BLDC/STEPPER FOC driver with NEMA17 footprint
TVS 48V
Supply voltage 8-48V (VM)
LDO 3.3V
DCDC 5V
Power Switch
FUSE 50mA
Power XH-2P
Magnetic Encoder MT6835 21bit
Control PH-6P
CAN PH-2P
Motor XH-4P
NEMA17
MCU STM32G431
Motor Driver with external MOSFETs
CAN-FD PHY Driver TJA1042
USB Type-C
ESD Protection
SWD
Power Daisy Chain XH2.5
Power Path, Reverse Polarity protection & 5V Sourcing for CTRL
5V Sourcing
Actual CTRL voltage is ~4.5V
Combined IO protection: Overcurrent at rated v; Overvoltage & ESD; Short Reverse voltage
Push-Pull or Open-Drain Logic
Compatible with 3.3V and 5V logic
Avoid reverse polarity
Speed up to 5MHz
Not to exceed 5V
Control IO protection
Control IO PullUp
Control connector
Do not use more than 50mA in 5V output mode
Title: CLN17 V3 Closed-Loop Motor Driver
More information on https://creapunk.com/cln17
Designed by CLN17 project
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CREAPUNK
Sheet: /
Sheet №: 1/1
Date: 2026-05
Rev: 1.1